

Precision STOPWATCH

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Most laboratories and educational institutions are in need of accurate stopwatches for precision time measurement. This simple DIY project focuses on the construction of a stopwatch with 0.01-second accuracy. The stopwatch can be used for sports events by connecting a sensor across connector CON2.

Circuit and working

The circuit diagram of the precision stopwatch is shown in Fig. 1. It is built around 7805 voltage regulator (IC1), AT89C4051 microcontroller (IC2), 4-digit common-anode 7-segment display (DIS1) and a few other components.

The circuit is powered from 230V AC mains via step-down transformer X1 (not shown here). X1 output (7.5V-

PARTS LIST

Semiconductors:

- | | |
|-------|--|
| IC1 | - 7805, 5V voltage regulator |
| IC2 | - AT89C4051 microcontroller |
| T1-T4 | - BC557 pnp transistor |
| D1-D2 | - 1N4007 rectifier diode |
| DIS1 | - 4-digit common-anode
7-segment display (KLQ56XX
or its equivalent) |

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|--------|----------------|
| R1-R7 | - 560-ohm |
| R8 | - 10-kilo-ohm |
| R9-R12 | - 5.6-kilo-ohm |

Capacitors:

- | | |
|--------|----------------------------------|
| C1 | - 1000 μ F, 25V electrolytic |
| C2, C3 | - 10nF, ceramic disc |
| C4 | - 100 μ F, 16V electrolytic |
| C5 | - 10 μ F, 16V electrolytic |
| C6, C7 | - 33pF, ceramic disc |

Miscellaneous:

- | | |
|--------|---|
| X1 | - 230V AC primary to
7.5V-0-7.5V, 500mA
secondary transformer |
| CON1 | - 3-pin connector |
| CON2 | - 2-pin connector |
| S1 | - On/off switch |
| S2, S3 | - Tactile switch |
| | - 20-pin IC base |
| XTAL1 | - 12MHz crystal |

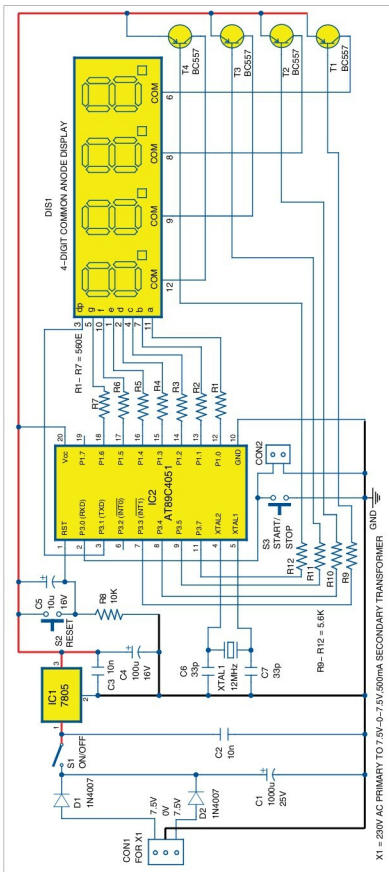


Fig. 1: Circuit diagram of the precision stopwatch